Synligare Overview

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Volvo Group, Advanced Technology and Research

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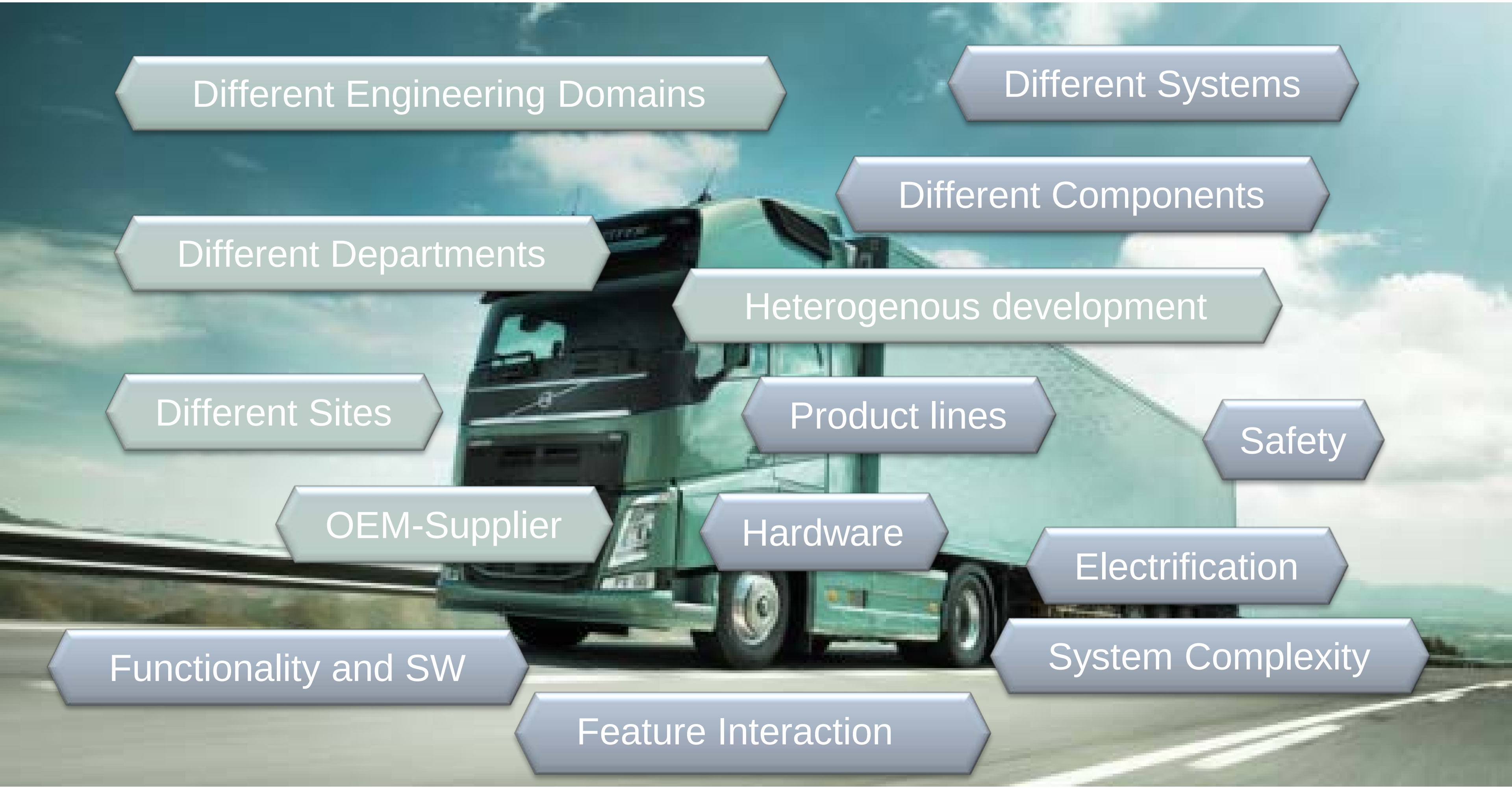
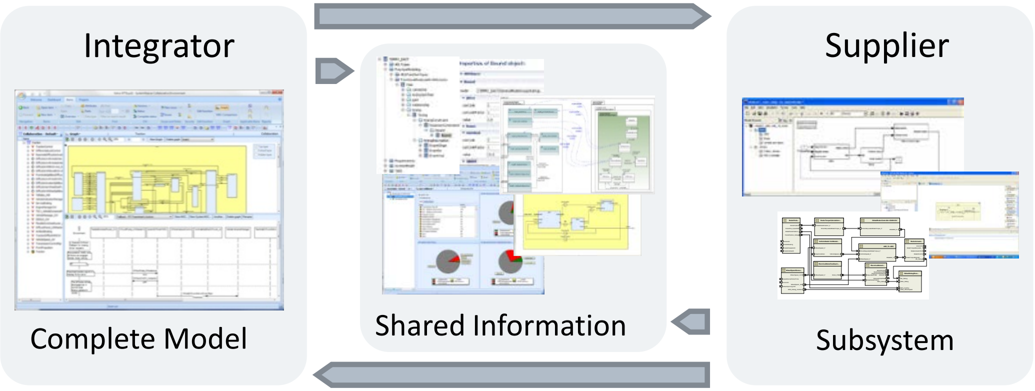
Process and Product Challenges



Research Activities - Synligare

- Goal:
  - Complexity-bridging cost-effective collaborative development
- Means:
  - Improved development methodology and tools through models
- Focus:
  - Supplier – OEM collaboration
  - Requirements management
  - Views and Metrics based on models



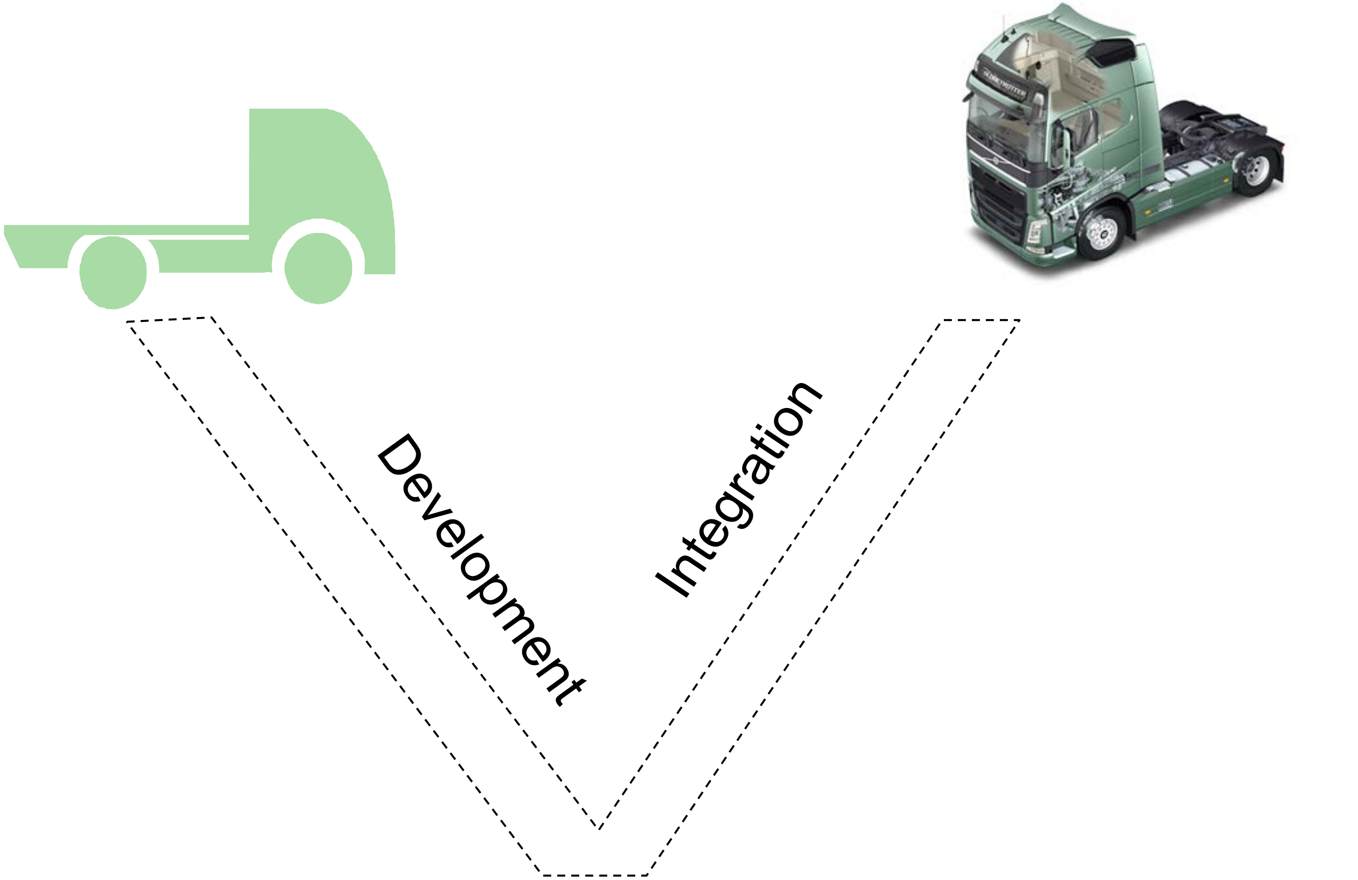
Synligare Project

- 2013-2016
- FFI Funding, 14 mio SEK
- OEM, Tier1, Tool Vendor, Engineering House
- Priniciples
  - Open source
  - Open standards
  - Information Efficiency
  - Collaboration Efficiency

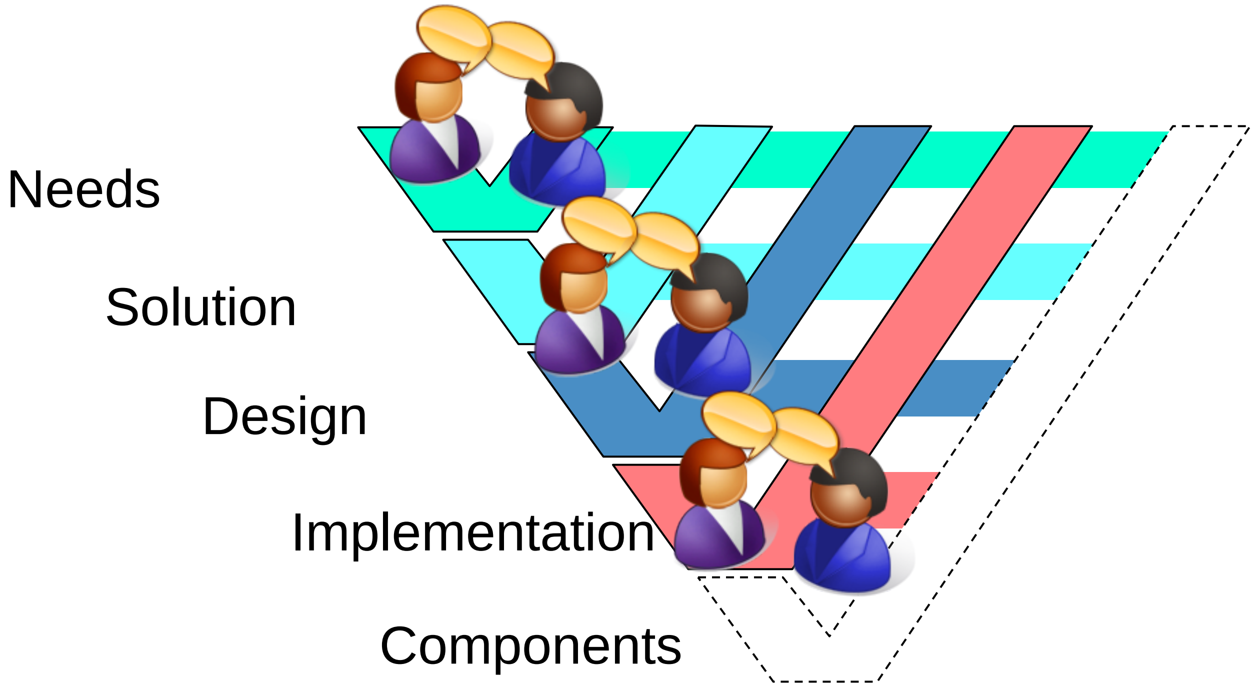
FUSE and HeavyRoad Challenges vs. Synligare Concepts

- Model Based Approach with precise representation
- Representation of
  - System Specification
    - Structure
    - Requirements
    - Timing
    - Safety
    - Energy Consumption
    - ...
- Model Exchange according to agreed formats and patterns
- Views and Metrics to help understanding information

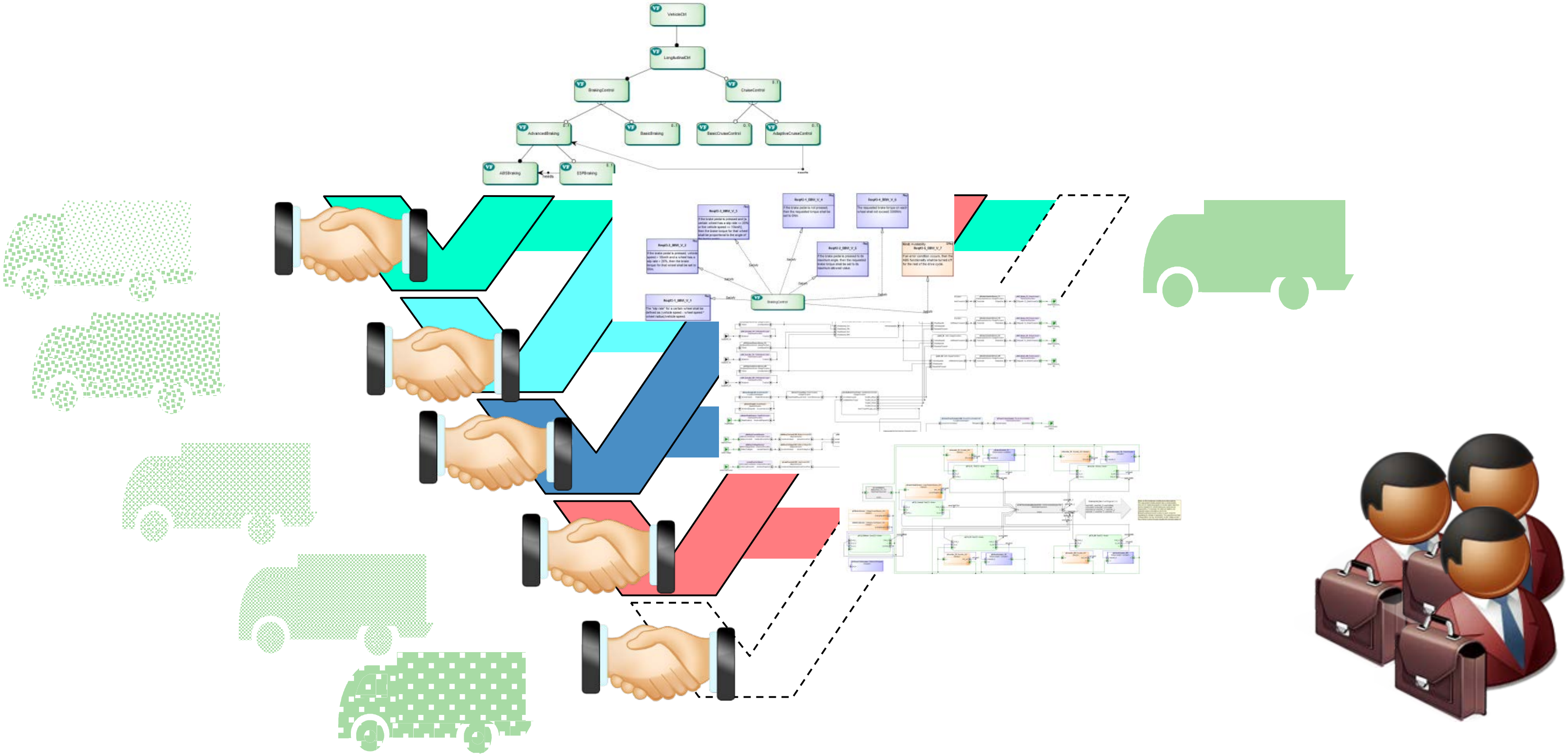
Engineering Life Cycle



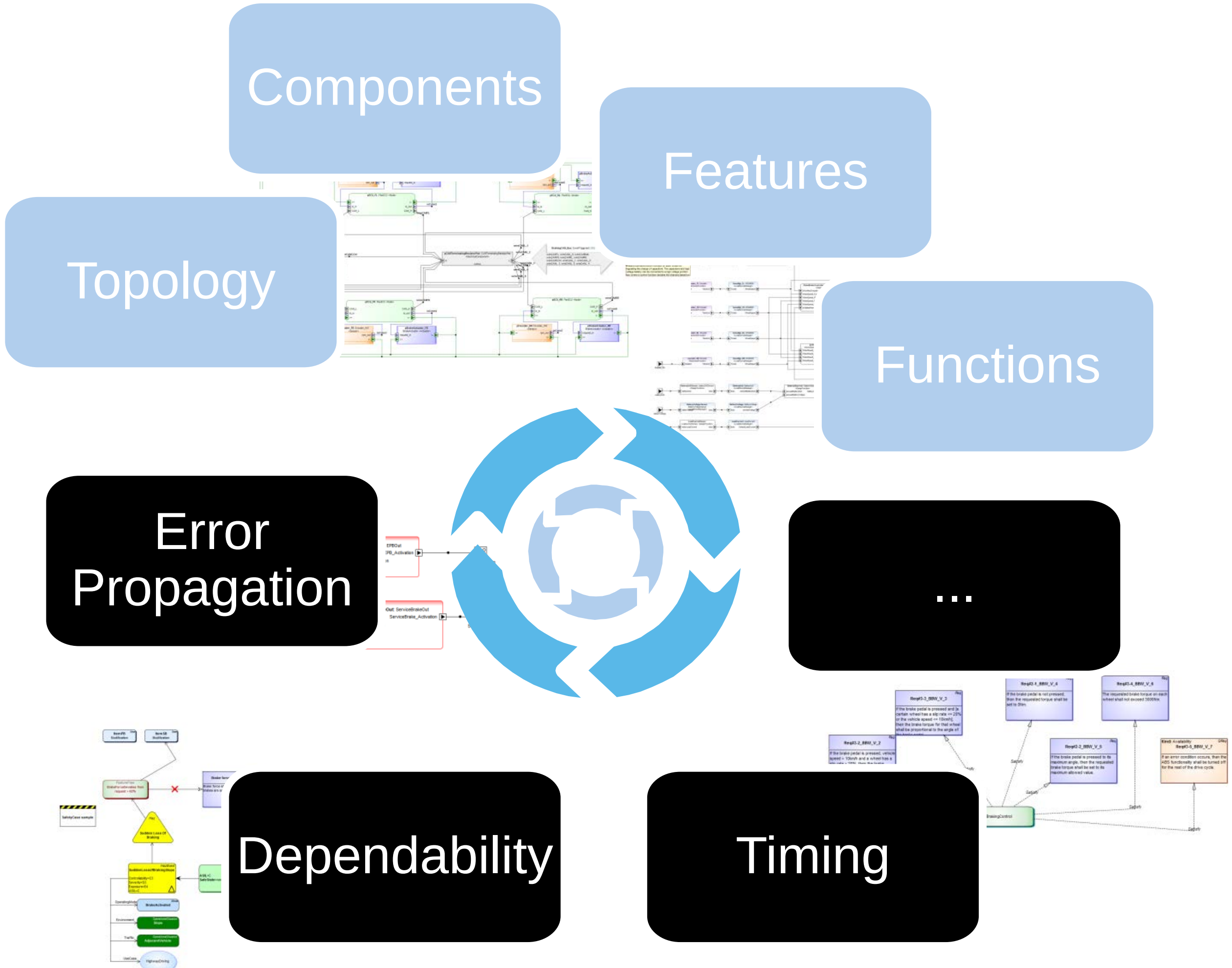
Engineering Life Cycle – Collaboration



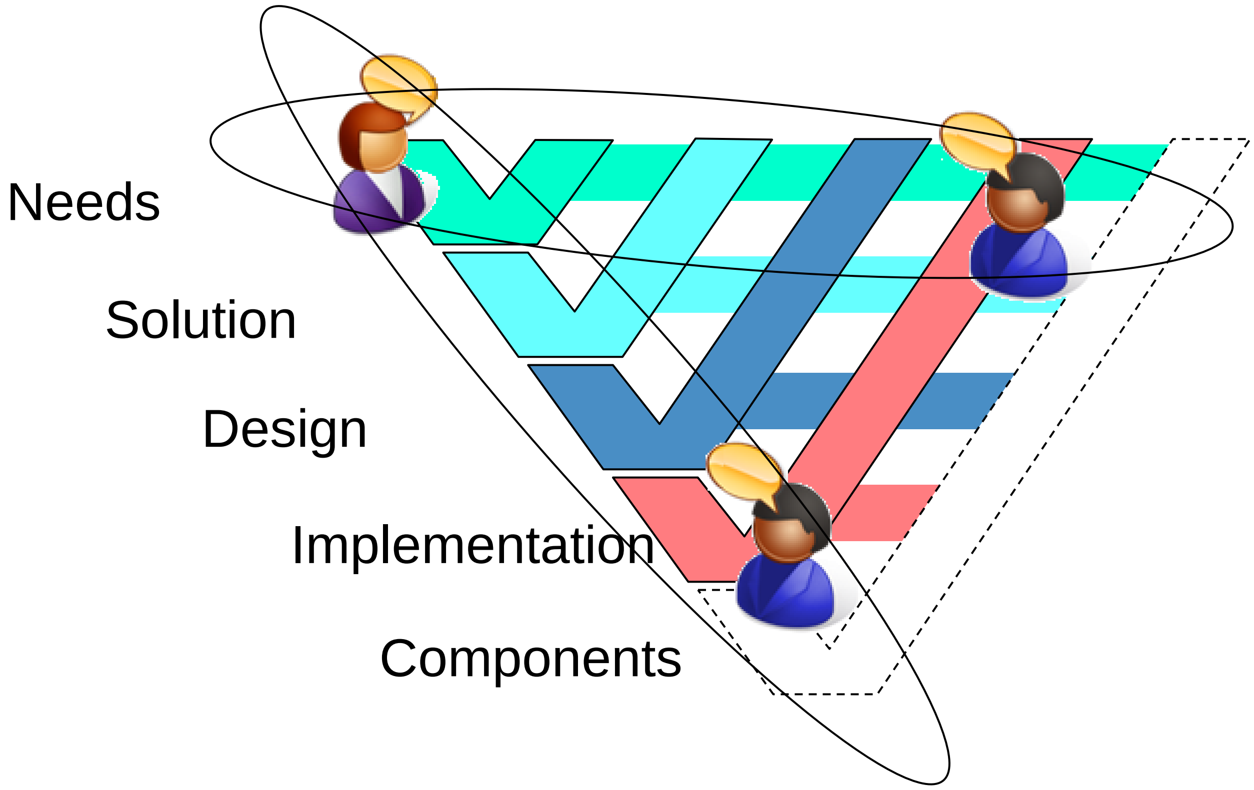
Engineering Life Cycle – Collaboration



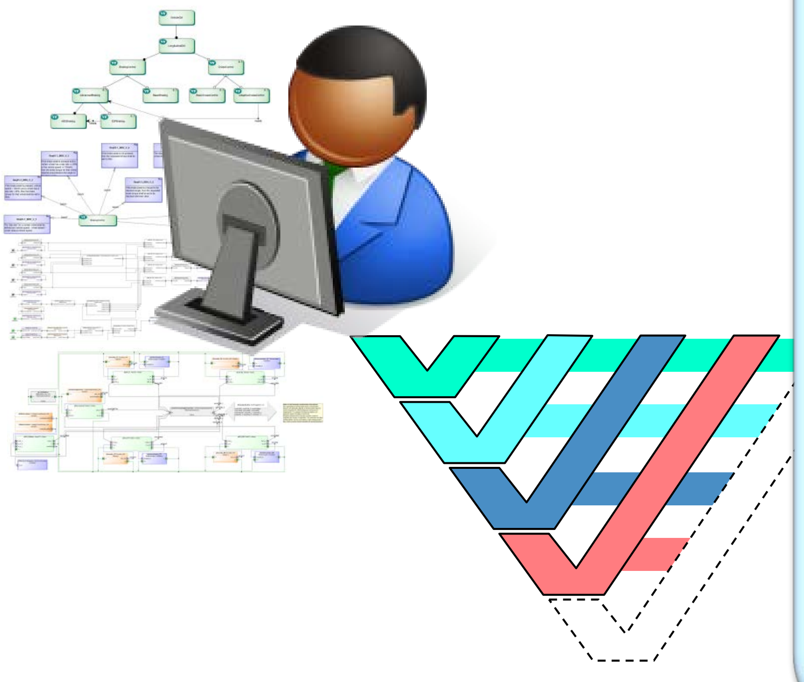
Model Based Engineering



Engineering Life Cycle – Iterations



Engineering Activities



Modelling and Synthesis

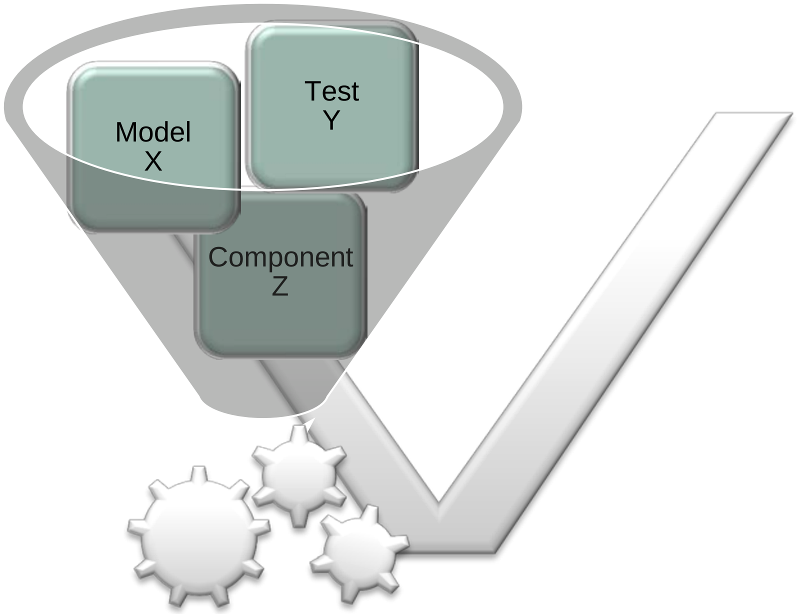
- Features
- Functions
- Topology
- Software Architecture
- Hardware Architecture
- Behavior
- Code
- ...

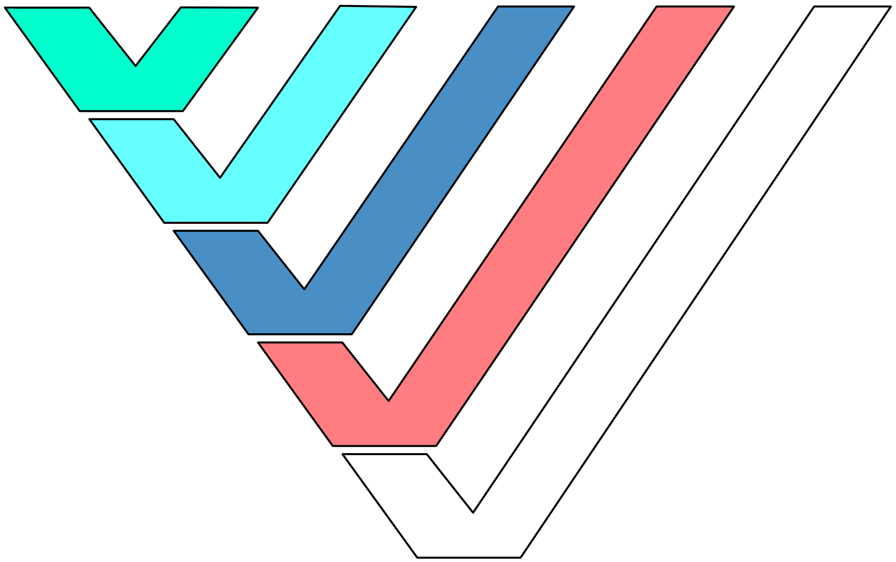
Analysis and Verification

- Timing
- Dependability
- Correctness
- Energy
- Cost
- ...

Continuous Integration and Delivery

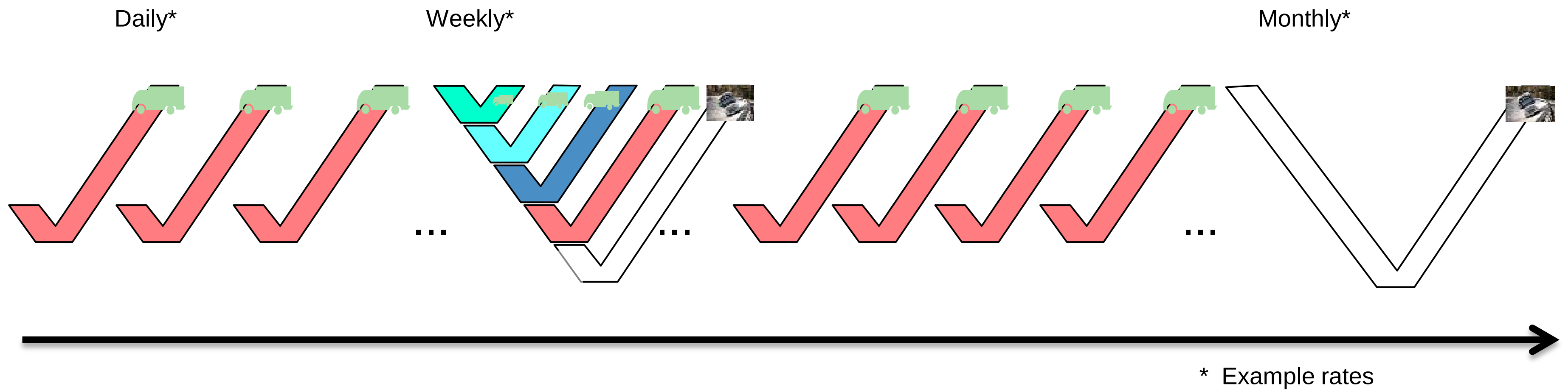
- Components provided to Build, Integration and Verification Server
  - Automatic integration based on system configuration
  - Automatic verification based on specified analyses or tests



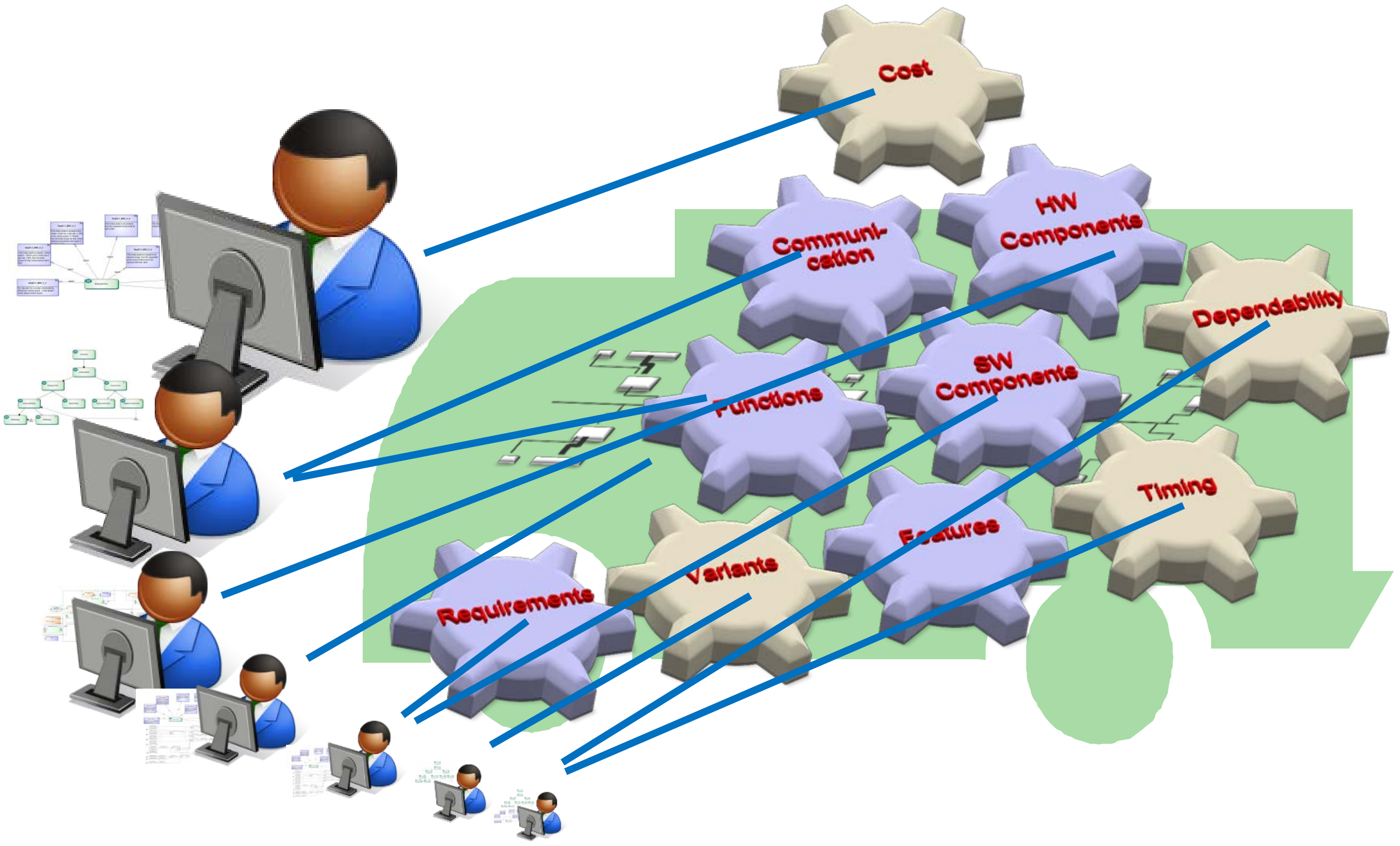


Continuous Integration and Delivery

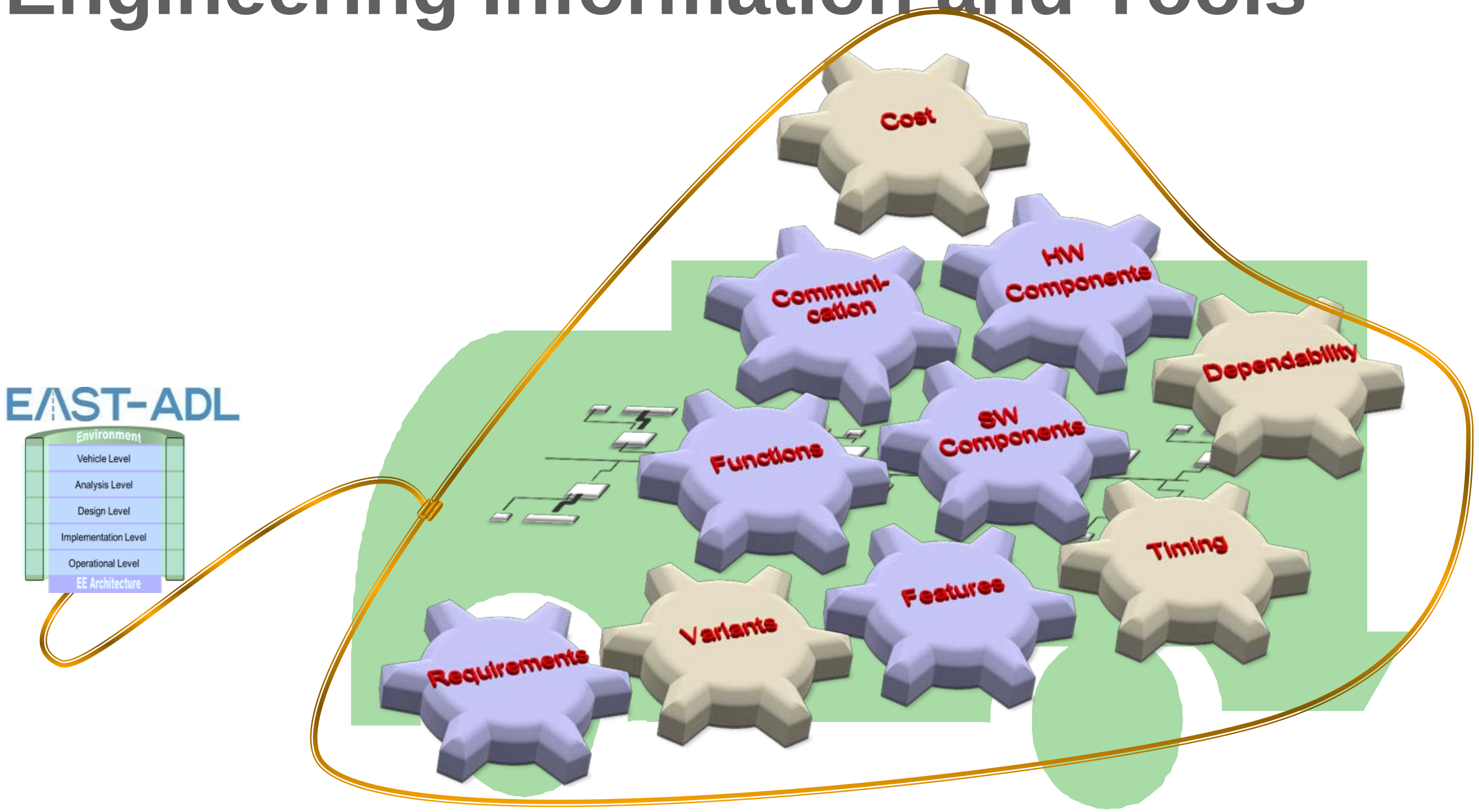
- Implementation and code refinements verified in frequent basic tests
- Design changes verified in elaborate analyses and tests
  - When implementation changes go beyond design envelope
  - Verification and validation of advanced objectives



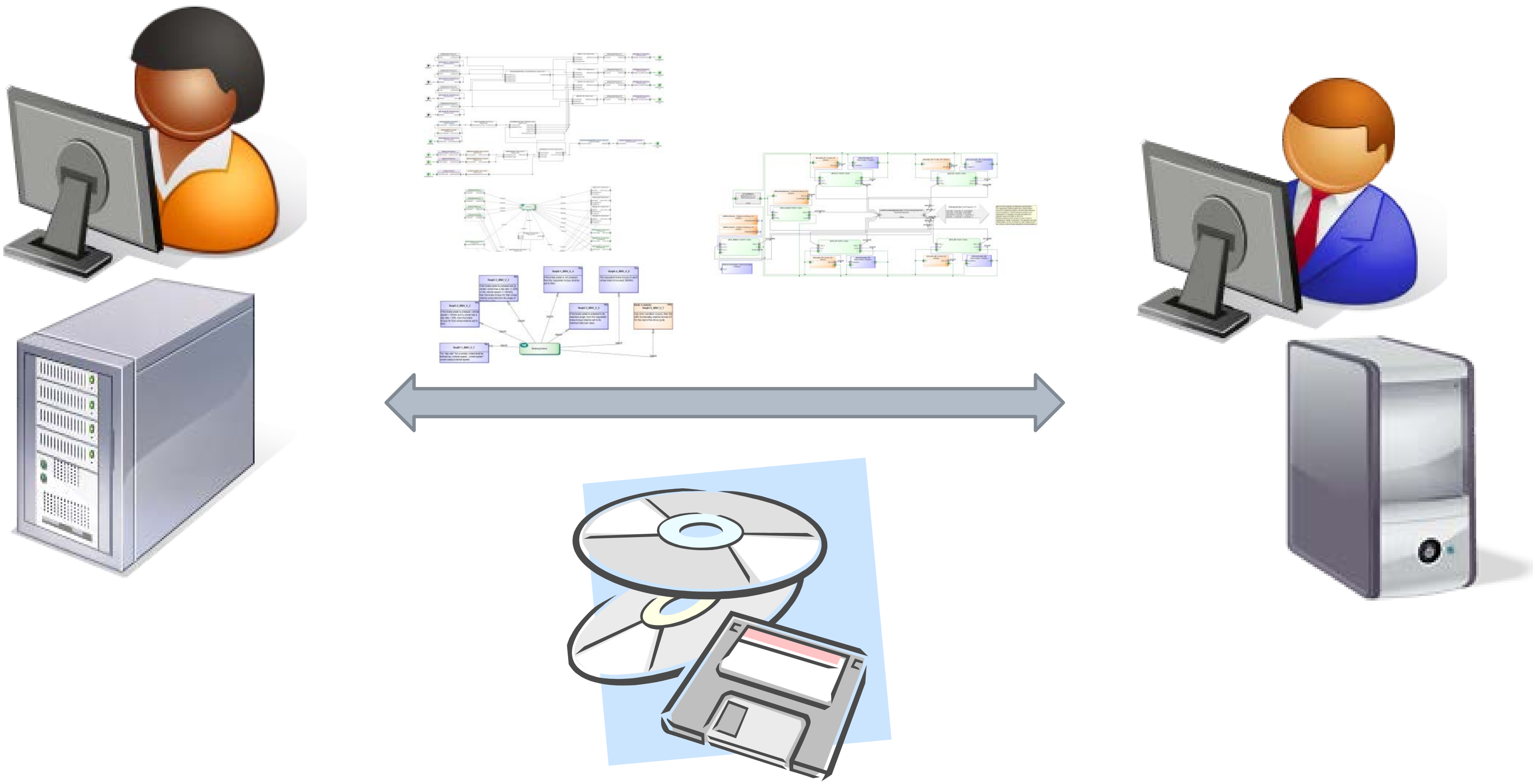
Engineering Information and Tools



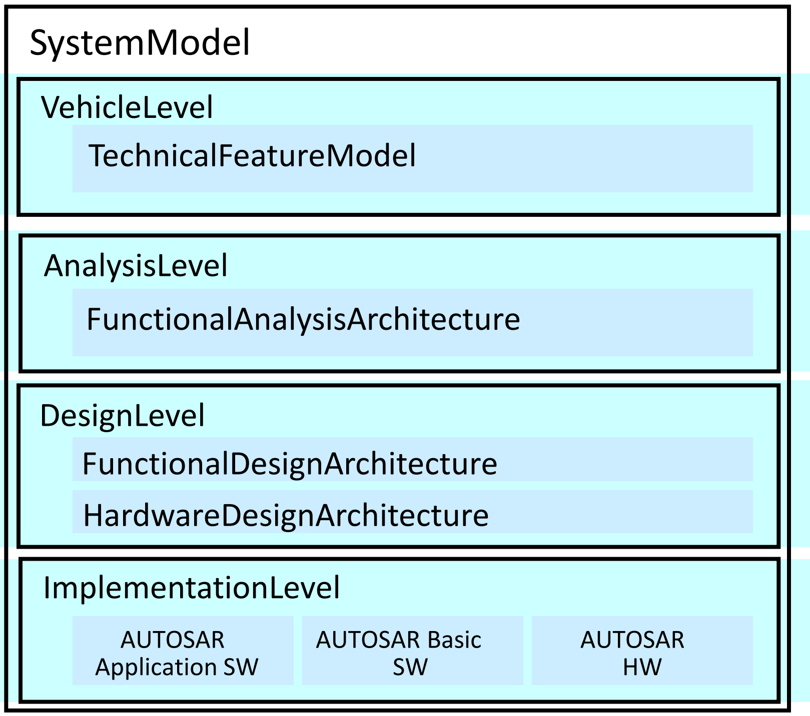
Engineering Information and Tools



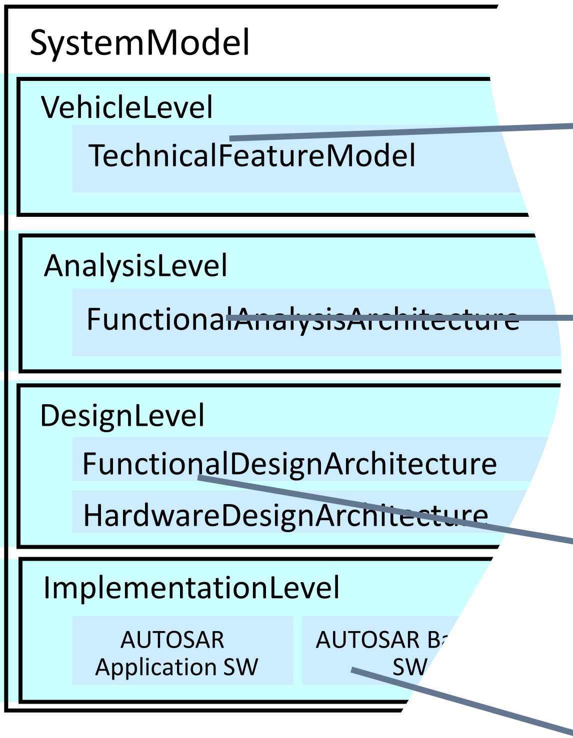
Data Exchange



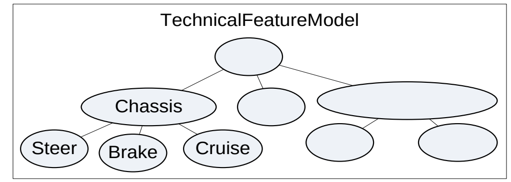
EAST-ADL Structure



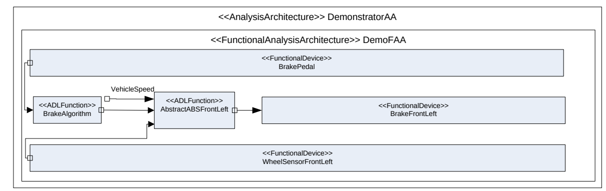
EAST-ADL Structure



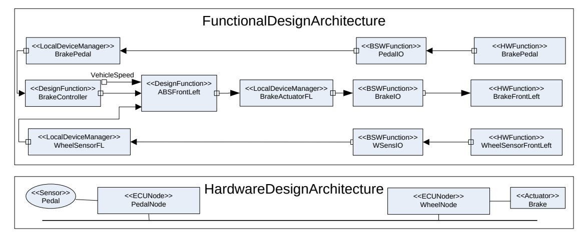
Features
of the vehicle



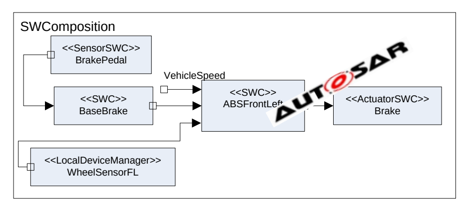
Abstract functions



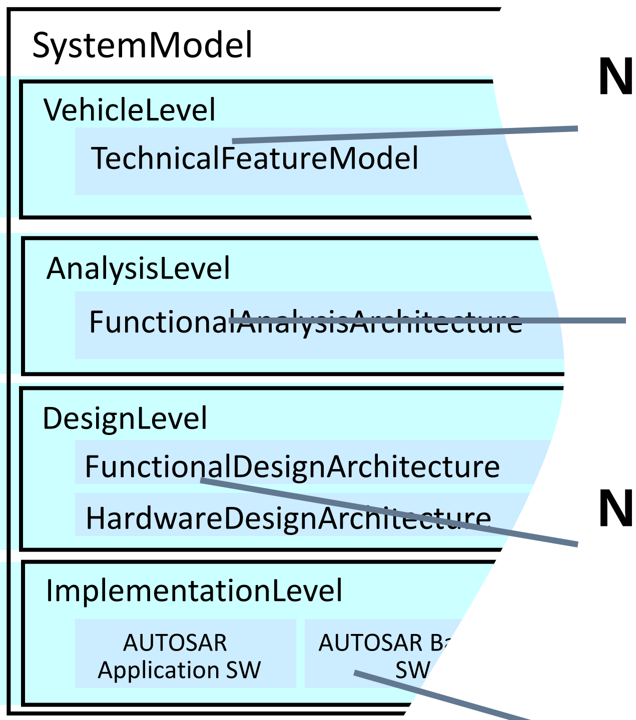
Hardware topology, concrete functions, allocation to nodes



Software Architecture
as represented
by AUTOSAR



EAST-ADL Structure



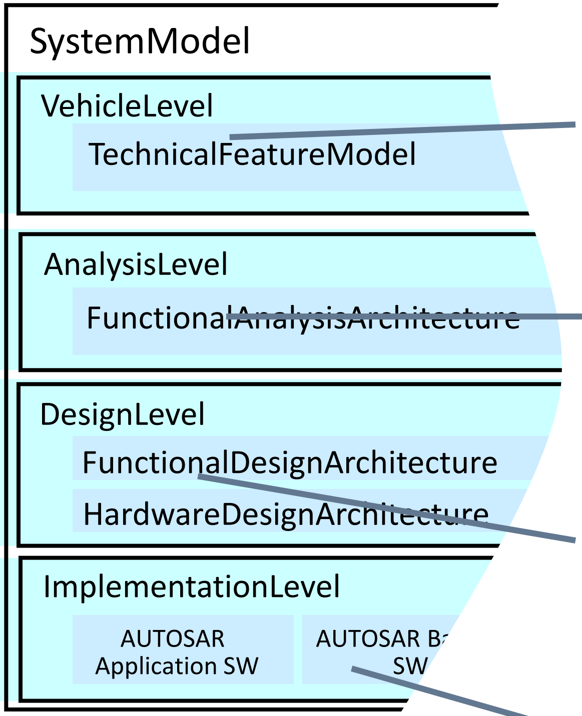
No Solutions

No Topology or Solution Details

No Software or Hardware Details

No Object Code, Silicon or Copper

EAST-ADL vs. ISO26262



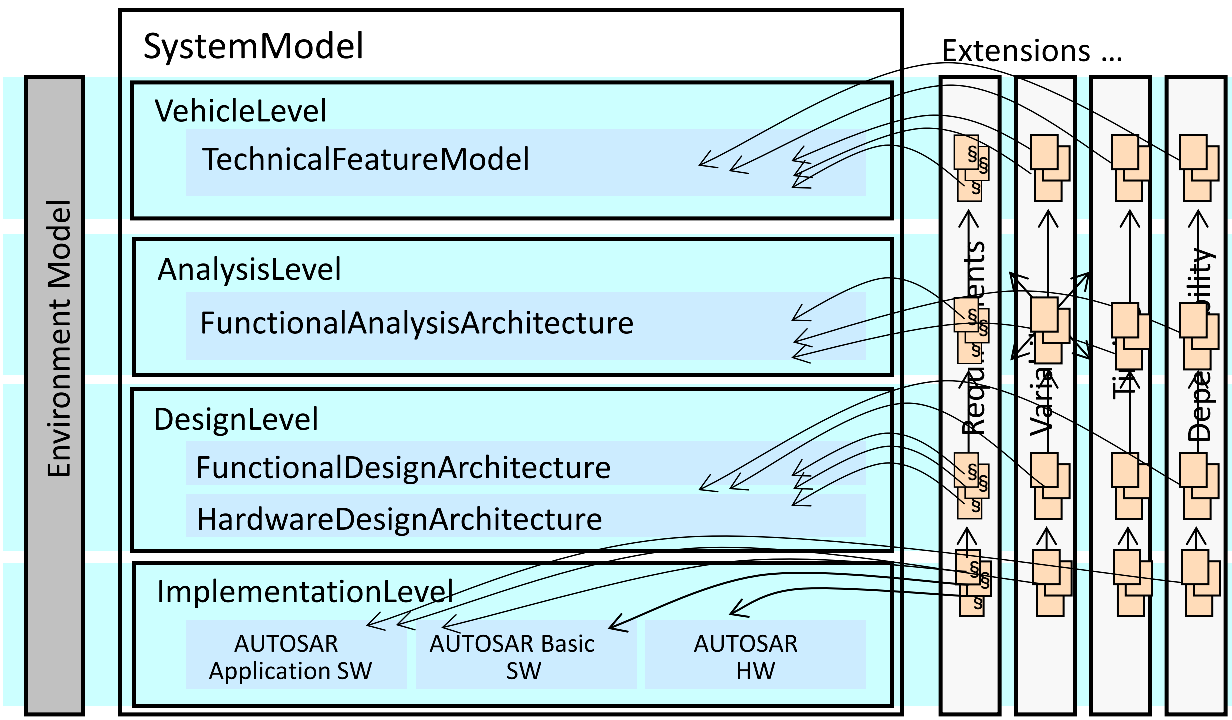
Focus on functional objectives and not technological solutions

Realization by high level architectural elements without notion of HW

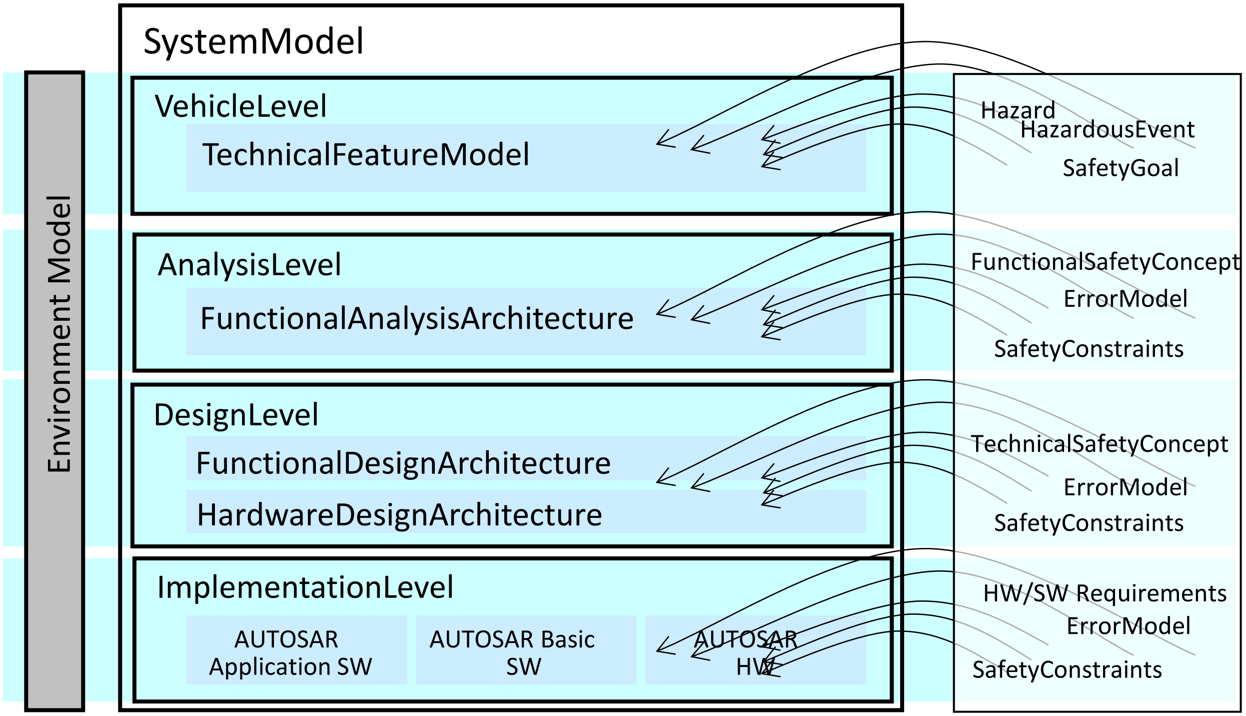
Introducing HW & SW in architecture

Implementation of SW/HW

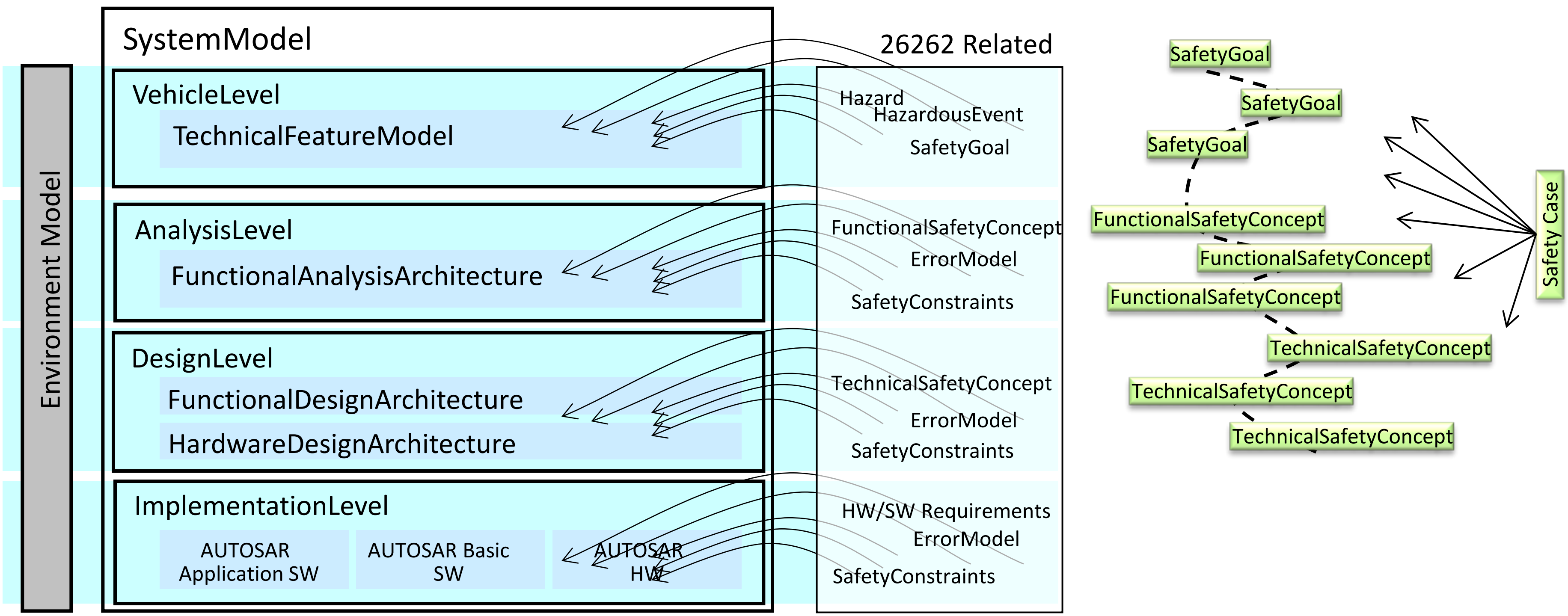
EAST-ADL Structure

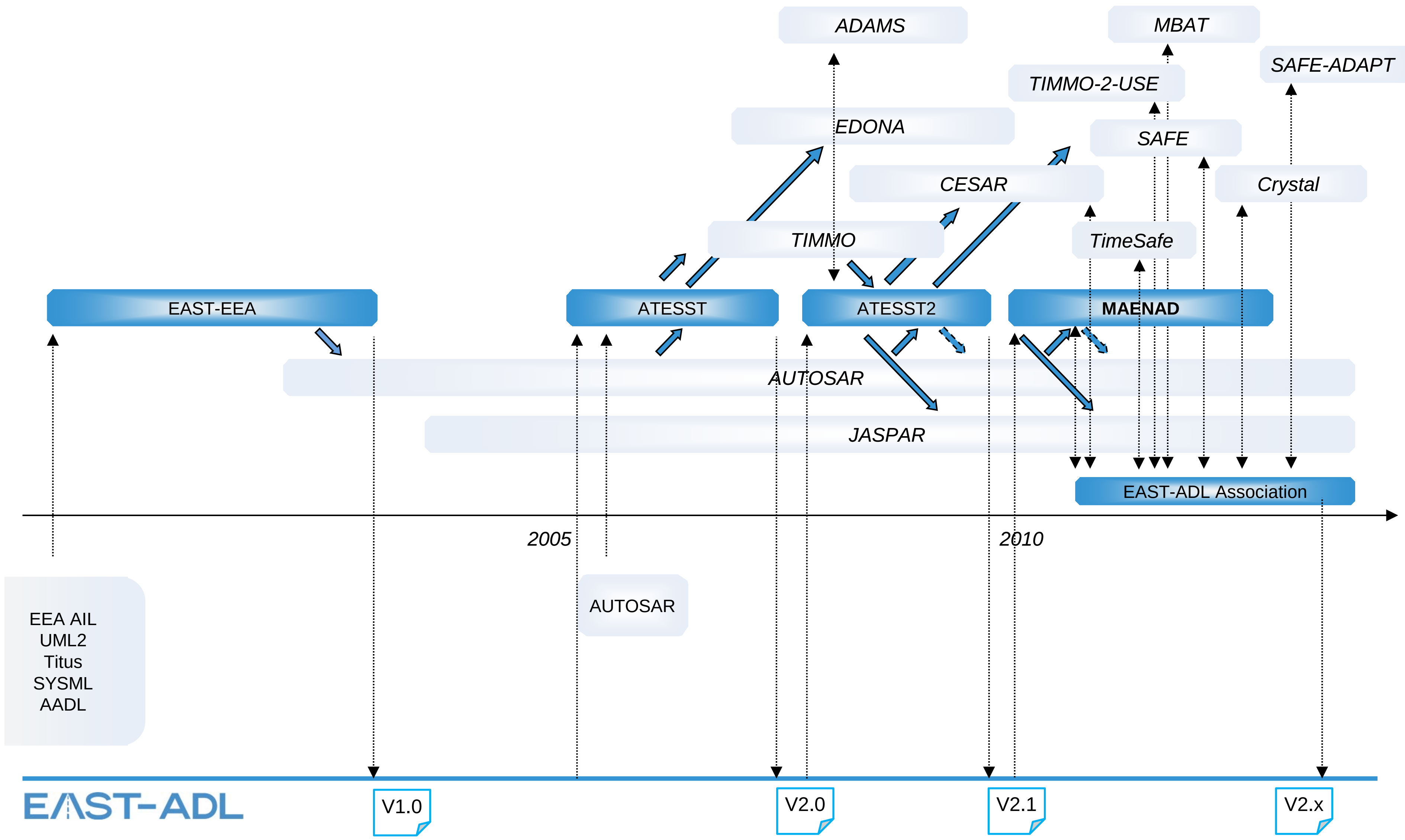


EAST-ADL Safety Artefacts



EAST-ADL Safety Artefacts

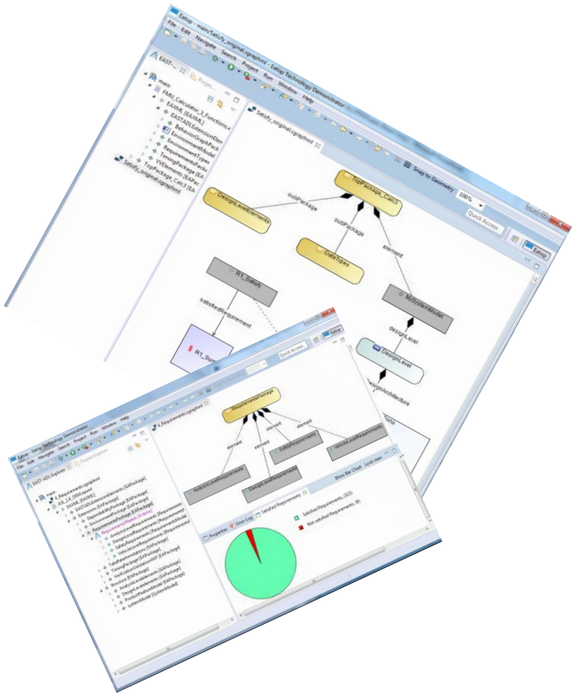



ADAMS
MBAT
SAFE-ADAPT
TIMMO-2-USE
EDONA
SAFE
CESAR
Crystal
TIMMO
TimeSafe
EAST-EEA
ATESST
ATESST2
MAENAD
AUTOSAR
JASPAR
EAST-ADL Association
2005
2010
EEA AIL
UML2
Titus
SYSML
AADL
AUTOSAR
EAST-ADL
V1.0
V2.0
V2.1
V2.x

EATOP - Eclipse-based Platform for EAST-ADL

- EAST-ADL Tool Platform
  - Open Source Platform for EAST-ADL tool development
  - Initiated by Eclipse Automotive Industry Working Group
    http://projects.eclipse.org/projects/modeling.eatop
  - Available at eclipse.org or synligare.eu
  - Follows ARTOP approach
    Based on same technology (Enterprise Architect MM, AUTOSAR modelling Rules, SPHINX, …)

Synligare Concepts



- Tools
  - SystemWeaver — systemite.se
  - EnterpriseArchitect — sparx.com
  - EATOP — eclipse.org
- Model Exchange based on EAST-ADL eaxml
- Views
  - Tables, Tracing, Tree browsing, Compare, Diagrams, Routing, ...
  - Graphical Exchange Format ”sgraphml”
- Metrics
  - Product and progress metrics
  - Generic approach based on xpath

Retrospective: Project Proposal

Projekt Synligare ämnar

- *ta fram metoder och verktyg för att effektivisera samarbetet mellan OEM och Tier-1 och därigenom ge långsiktig förmåga att utveckla komplexa inbyggda programvarusystem för fordon.* √
- *Identifiera en representation för systembeskrivningar med fokus på kravinformation och dess koppling till systemkomponenter.* √
- *Identifiera relevanta vyer, nyckeltal och analyser till stöd för nedbrytning, realisering och uppföljning av krav.* √
- *Implementera experimentella verktygskedjor för validering av framtagna koncept.* √
- *Validera metodik i ett tänkt OEM-Tier-1-samarbete* √

Retrospective: Project Proposal

OEM utvecklar fordon och preliminära versioner av ingående system

OEM beställer en komponent genom att kommunicera en delmängd av systembeskrivningen, strukturerad enligt en utbytesmodell

Tier-1 granskar samt kompletterar med ytterligare krav och egenskaper

Tier-1 kommunicerar komponentspecifikation, strukturerad enligt utbytesmodell

OEM integrerar och förfinar specifikationerna av ingående system

OEM kommunicerar slutlig komponentspecifikation/modell

Iterationer

Tier-1 utvecklar komponent

Tier-1 levererar komponenten till OEM med aktuell specifikation

OEM integrerar specifikationerna av ingående system

OEM kommunicerar uppdateringar och korrigeringar

Systembeskrivningen tillåter att visualiseringar, analyser och mätetal genereras på utbytt information

WWW.EAST-ADL.INFO

EASTADL

www.synligare.eu

Thank You for Your Attention!

.... Now let's look at an example scenario